3.1 Introduction

Molecules of alkyne series of hydrocarbon are characterized by having two adjacent carbon atoms joined to one another by a triple bond. The carbon-carbon triple bond is unsaturated and highly reactive toward the reagents that double bonds react with.

The general formula is $\mathbf{C_nH_{2n-2}}$.

$HC \equiv CH$	Ethyne (acetylene)
$HC \equiv CCH_3$	Propyne
$HC \equiv CCH_2CH_3$	1-Butyne
$HC \equiv C(CH_2)_2CH_3$	1-Pentyne
$HC \equiv C(CH_2)_3CH_3$	1-Hexyne
$HC \equiv C(CH_2)_4CH_3$	1-Heptyne
$HC \equiv C(CH_2)_5CH_3$	1-Octyne
$HC \equiv C(CH_2)_6CH_3$	1-Nonyne
$HC \equiv C(CH_2)_7CH_3$	1-Decyne

3.2 Nomenclature

The alkynes are named according to two systems. In one, they are considered to be derived from acetylene by replacement of one or both hydrogen atoms by alkyl group.

$H - C \equiv C - C_2H_5$ (Ethylacety ٤١ من ٥٦ **tyne**

$CH_3 - C \equiv C - CH_3$ (Dimethylacetylene) 2-Butyne

$CH_3 - C \equiv C - CH(CH_3)_2$ (Isopropylmethylacetylene) 4-Methyl-2-pentyne

$HC \equiv CH$	Ethyne (acetylene)
$HC \equiv CCH_3$	Propyne
$HC \equiv CCH_2CH_3$	1-Butyne
$HC \equiv C(CH_2)_2CH_3$	1-Pentyne
$HC \equiv C(CH_2)_3CH_3$	1-Hexyne
$HC \equiv C(CH_2)_4CH_3$	1-Heptyne
$HC \equiv C(CH_2)_5CH_3$	1-Octyne
$HC \equiv C(CH_2)_6CH_3$	1-Nonyne
$HC \equiv C(CH_2)_7CH_3$	1-Decyne

3.2 Nomenclature

The alkynes are named according to two systems. In one, they are considered to be derived from acetylene by replacement of one or both hydrogen atoms by alkyl group.

$\mathbf{H - C \equiv C - C_2H_5}$	**(Ethylacetylene) 1-Butyne**
$\mathbf{CH_3 - C \equiv C - CH_3}$	**(Dimethylacetylene) 2-Butyne**
$\mathbf{CH_3 - C \equiv C - CH(CH_3)_2}$	**(Isopropylmethylacetylene) 4-Methyl-2-pentyne**

For more complicated alkyne the IUPAC names are used. The rules are exactly as for the alkenes, except that the ending **-yne** is used in place of **–ene**.

$CH_3C{\equiv}CCH_2CH_3$ — $CH_3CH{=}CHCH_2C{\equiv}CH$

2-Pentyne — 4-Hexen-1-yne

$CH_3CH(CH_3)CH_2C{\equiv}CCH_2CH_3$ — $CH_3C{\equiv}CCH_2CH(CH_3)CH_2CH_3$

6-Methyl-3-heptyne — 5-Methyl-2-heptyne

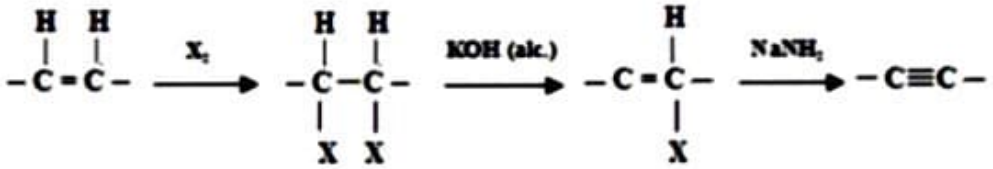
3.3 Physical Properties of Alkynes

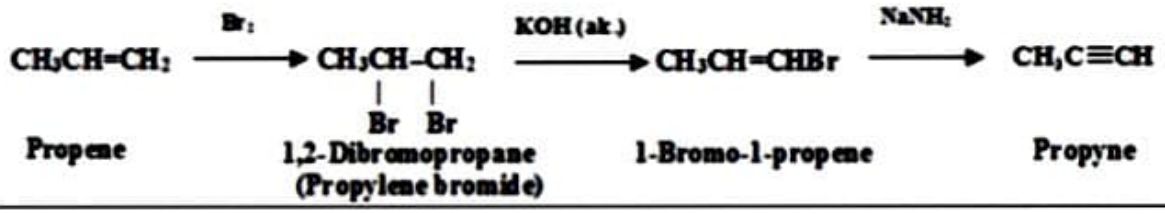
Being compounds of low polarity, the alkynes have physical properties that are essentially the same as those of the alkanes and alkenes. They are insoluble in water but quite soluble in the usual organic solvents of low polarity: ether, benzene, and carbon tetrachloride. They are less dense than water. Their boiling points show the usual increase with increasing carbon number, and the usual effects of chain branching; they are very nearly the same as the boiling points of alkanes and alkenes with the same carbon skeletons.

3.4 Preparation of Alkynes

1- Dehydrohalogenation of alkyl halides

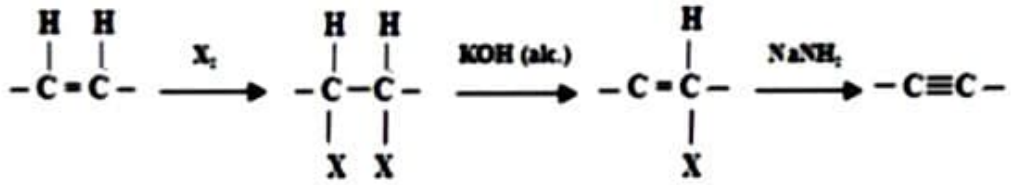
A carbon-carbon triple bond is formed in the same way as a double bond: elimination of atoms or groups from two adjacent carbons.

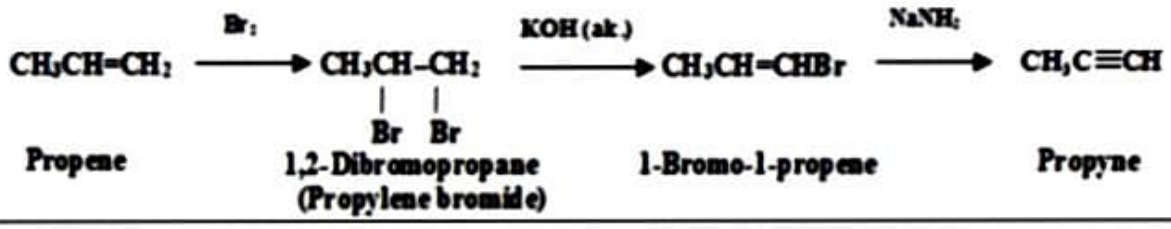




1- Dehydrohalogenation of alkyl halides

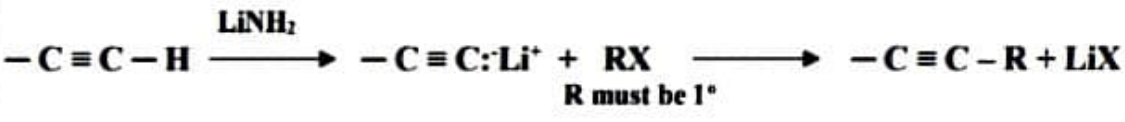
A carbon-carbon triple bond is formed in the same way as a double bond: elimination of atoms or groups from two adjacent carbons.

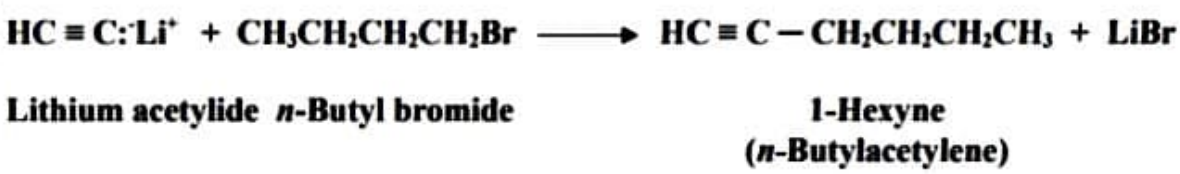




2- Reaction of metal acetylides with alkyl halides

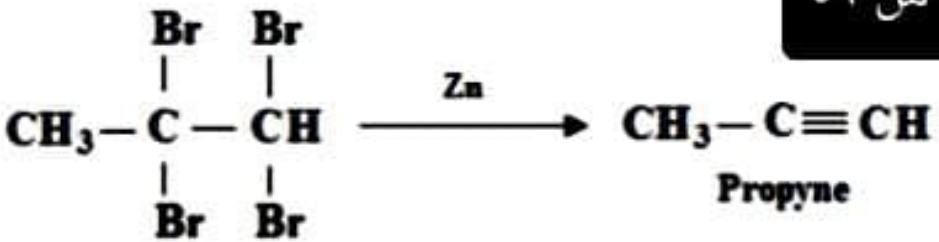
Lithium or sodium acetylides can react with primary alkyl halides. The alkyl group becomes attached to the triply bonded carbon, and a new, larger alkyne has been generated.





3- Dehalogenation of tetrahalides

Alkynes can be prepared by dehalogenation of tetrahalides.



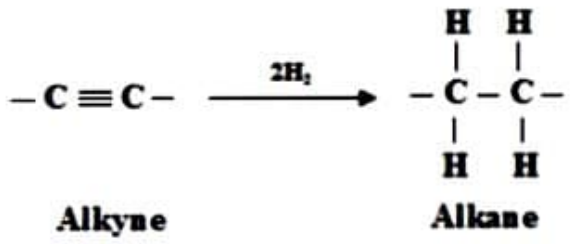
3.5 Reactions of Alkynes

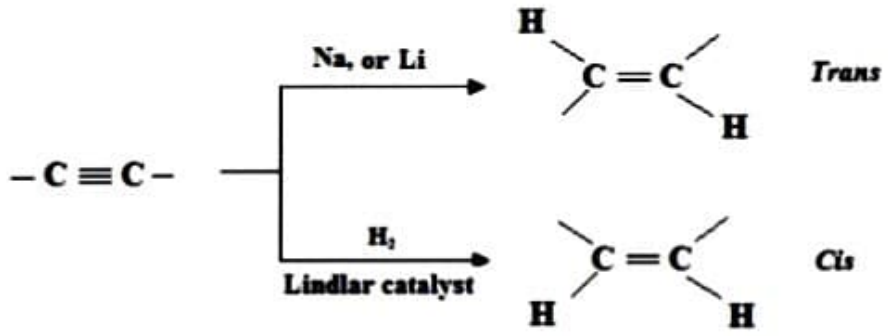
The carbon-carbon triple bond –C≡C– consists of one σ-bond and two weaker π-bonds, so it is not surprising that addition reactions are a characteristic feature of this functional group. Despite its high formal unsaturation, the triple bond dose not usually react so vigorously as a carbon-carbon double bond, as in alkenes, and it is therefore sometimes possible to stop the addition reaction after the uptake of 1 mol of reactant, rather than 2 mol which would be possible in principle.

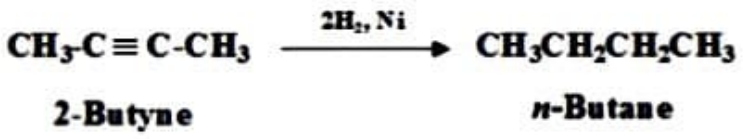
(Addition Reactions)

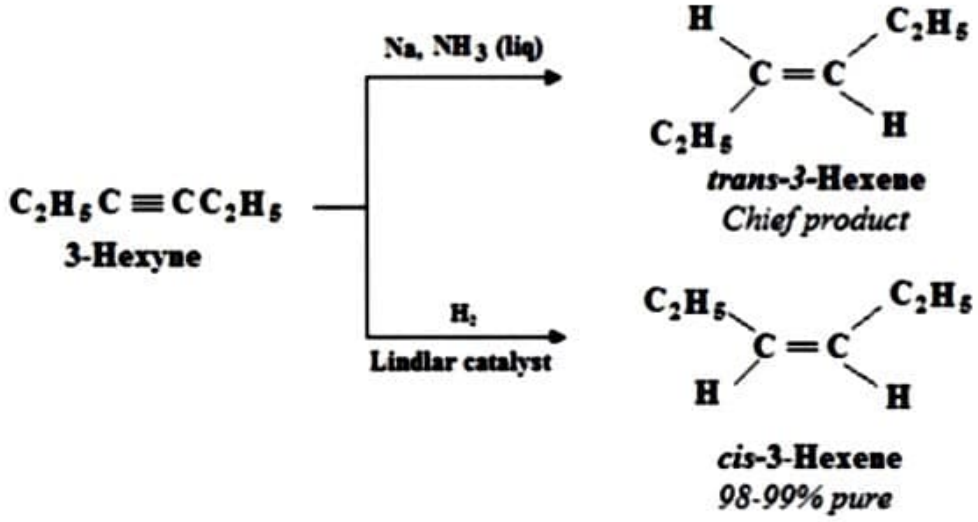
1- Addition of hydrogen

As would be expected, alkynes can be hydrogenated catalytically, taking up 2 mol of hydrogen per mol of alkyne to form the corresponding saturated molecule. Careful control of the hydrogenation can stop the reaction at the alkene stage.



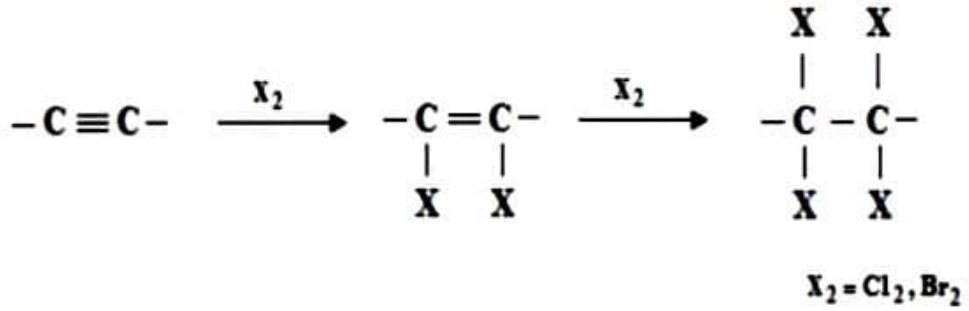


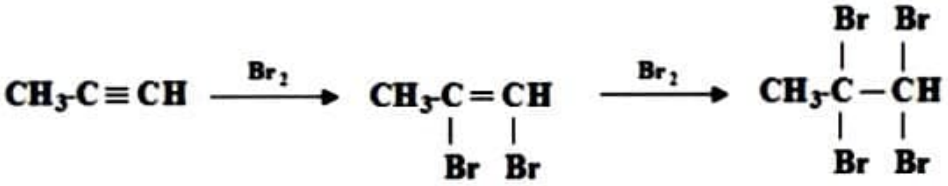




2- Addition of halogen

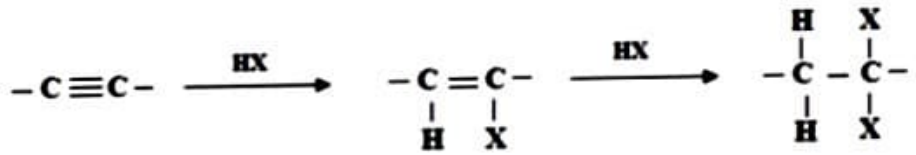
Alkynes can react with chlorine and bromine to yield tetrahaloalkanes. A dihaloalkene is an intermediate.



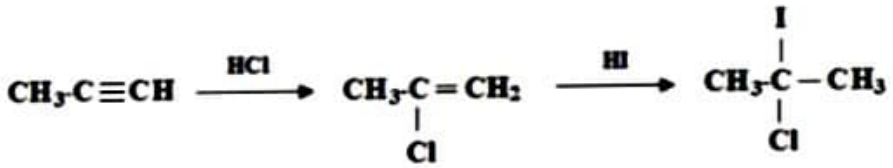


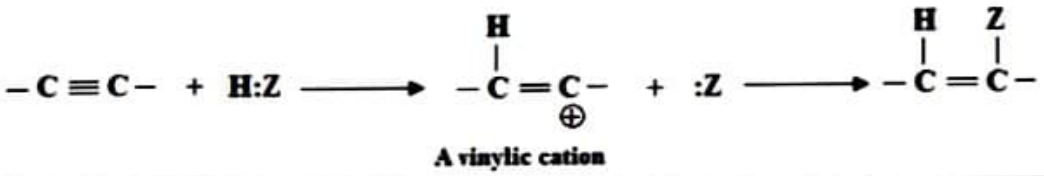
3- Addition of hydrogen halides

Alkynes also react with hydrogen halides to form alkyl halides. Addition of acids like hydrogen halides is electrophilic addition, and it appears to follow the same mechanism with alkynes as with alkenes: via an intermediate carbocation. The difference is that here the intermediate is a *vinylic cation*.



HX = HCl, HBr, HI

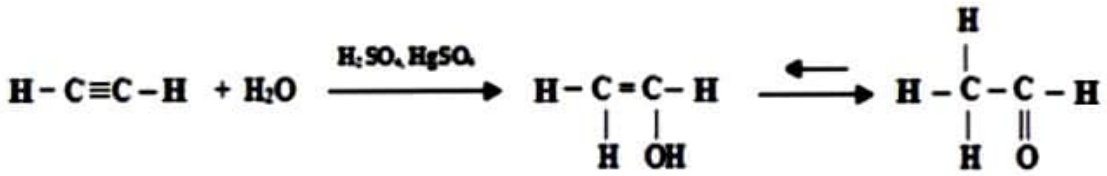




A vinylic cation

4- Addition of water (hydration)

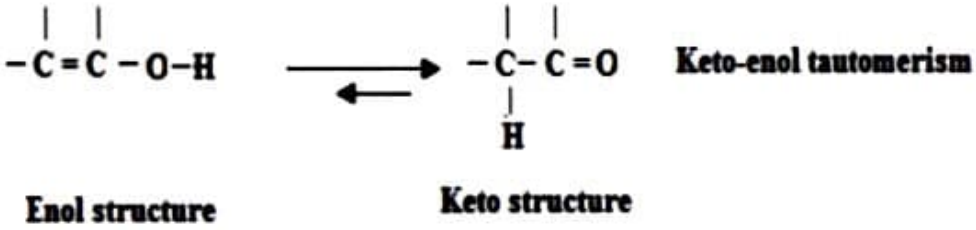
In the presence of dilute sulphuric acid and a mercury(II) sulphate catalyst at about 60°C, water adds across a triple bond and alkyne is said to be hydrated.

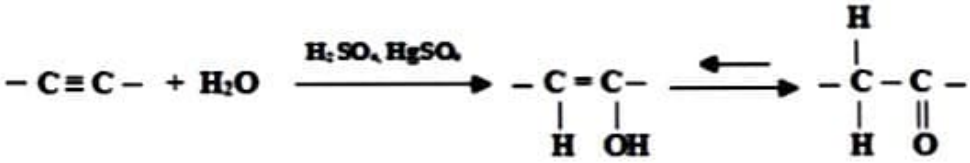


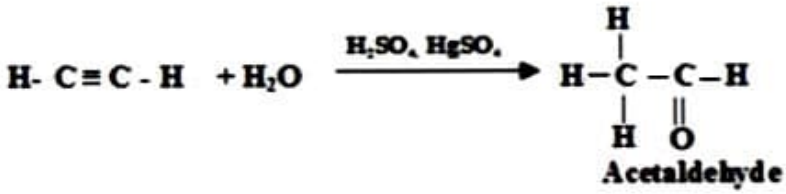
Acetylene — Vinyl alcohol — Acetaldehyde

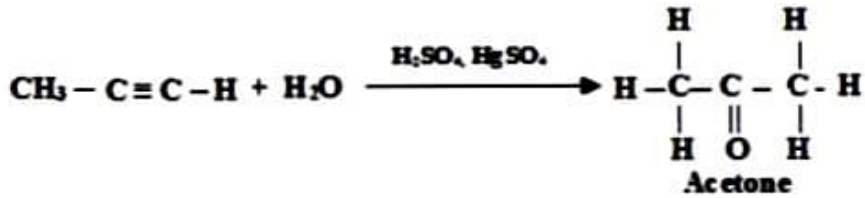
If hydration of acetylene followed the same pattern as hydration of alkenes, we would expect addition of H- and –OH to the triple bond to yield *vinyl alcohol*. But all attempts to prepare vinyl alcohol result, like hydration of acetylene, in the formation of acetaldehyde.

A structure with –OH attached to doubly bonded carbon is called **enol** (*-ene* for the carbon-carbon double bond, *-ol* for *alcohol*). It is almost always true that when we try to make a compound with the enol structure, we obtain instead a compound with **keto**structure (one that contains a C=O group). There is an equilibrium between the two structures, but it generally lies very much in favor of the keto form.





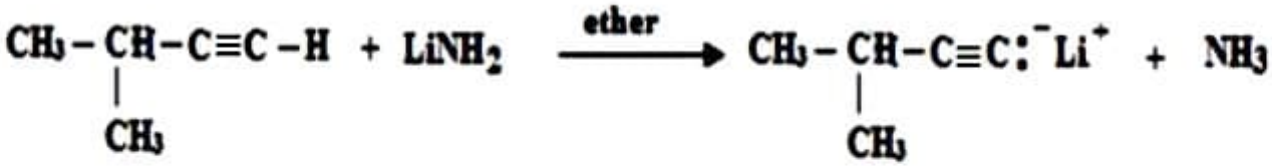




$$-C{\equiv}C-H + \text{base} \longrightarrow -C{\equiv}C:^{-}$$

Formation of metal acetylides

Hydrogen attached to triply bonded carbon, as in acetylene or any alkyne with the triple bond at the end of the chain (RC≡C-H), shows appreciable acidity. So when a terminal triple bond is treated with an alkylmagnesium halide or an alkyllithium, the alkane is displaced from its salt, and the metal acetylide is obtained.



Lithium isopropylacetylide

Lithium methyacetylide

Ethynylmagnesium bromide

Heats of Hydrogenation

$CH_3CH_2C{\equiv}CH$

292 kJ/mol

$CH_3C{\equiv}CCH_3$

275 kJ/mol

Alkyl groups stabilize triple bonds in the same way that they stabilize double bonds. Internal triple bonds are more stable than terminal ones.

Partial Hydrogenation

$$RC{\equiv}CR' \xrightarrow[\text{cat}]{H_2} RCH{=}CHR' \xrightarrow[\text{cat}]{H_2} RCH_2CH_2R'$$

- Alkynes could be used to prepare alkenes if a catalyst was available that is active enough to catalyze the hydrogenation of alkynes, but not active enough for the hydrogenation of alkenes.

Lindlar Palladium

$$RC{\equiv}CR' \xrightarrow[\text{cat}]{H_2} RCH{=}CHR' \xrightarrow[\text{cat}]{H_2} RCH_2CH_2R'$$

- There is a catalyst that will catalyze the hydrogenation of alkynes to alkenes, but not that of alkenes to alkanes.
- It is called the Lindlar catalyst and consists of palladium supported on $CaCO_3$, which has been poisoned with lead acetate and quinoline.
- *syn*-Hydrogenation occurs; cis alkenes are formed.

Alkynes. C_nH_{2n-2}

C_2H_2 H:C:::C:H H—C ≡ C—H sp => linear, 180°

acetylene

ethyne

C_3H_4 $CH_3C{\equiv}CH$ methylacetylene

propyne

nomenclature:

common names: "alkylacetylene"

IUPAC: parent chain = longest continuous carbon chain that contains the triple bond.

alkane drop –ane add -yne

prefix locant for the triple bond, etc.

$CH_3CH_2C{\equiv}CCH_3$ 2-pentyne

ethylmethylacetylene

"terminal" alkynes have the triple bond at the end of the chain:

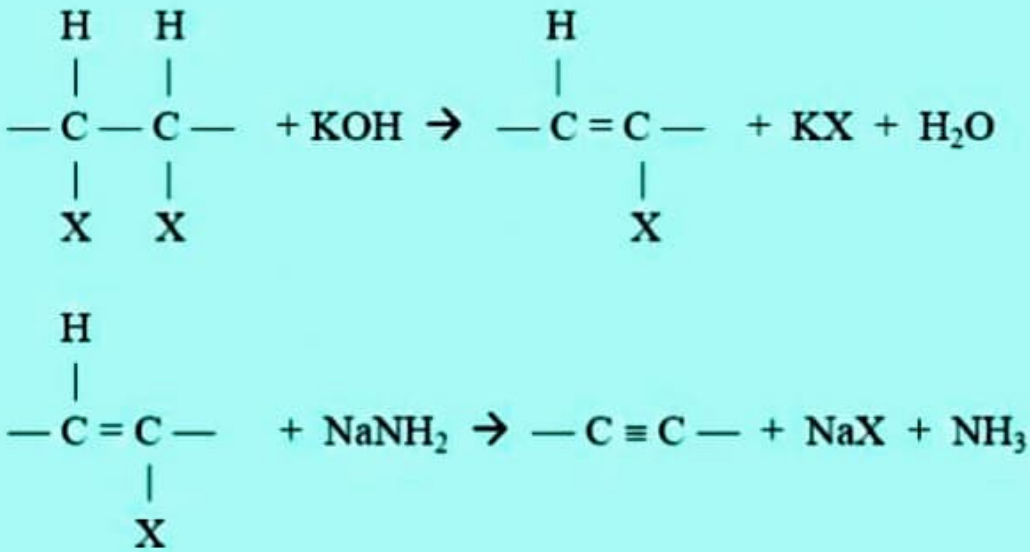
$CH_3CH_2C{\equiv}CH$	$HC{\equiv}CCH(CH_3)CH_2CH_3$
1-butyne	3-methyl-1-pentyne
ethylacetylene	*sec*-butylacetylene

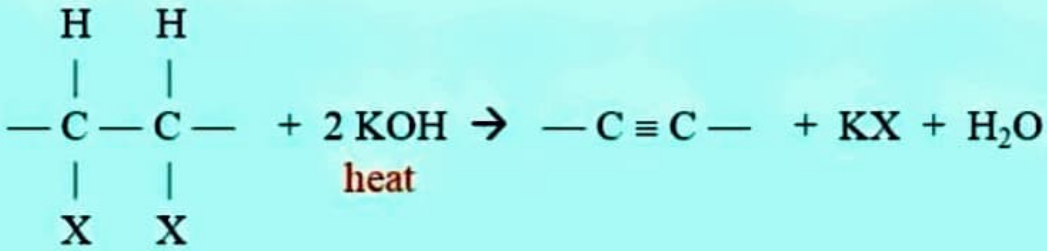
physical properties:

- weakly or non-polar, no H-bonding
- relatively low mp/bp
- water insoluble

Synthesis, alkynes:

1. dehydrohalogenation of vicinal dihalides





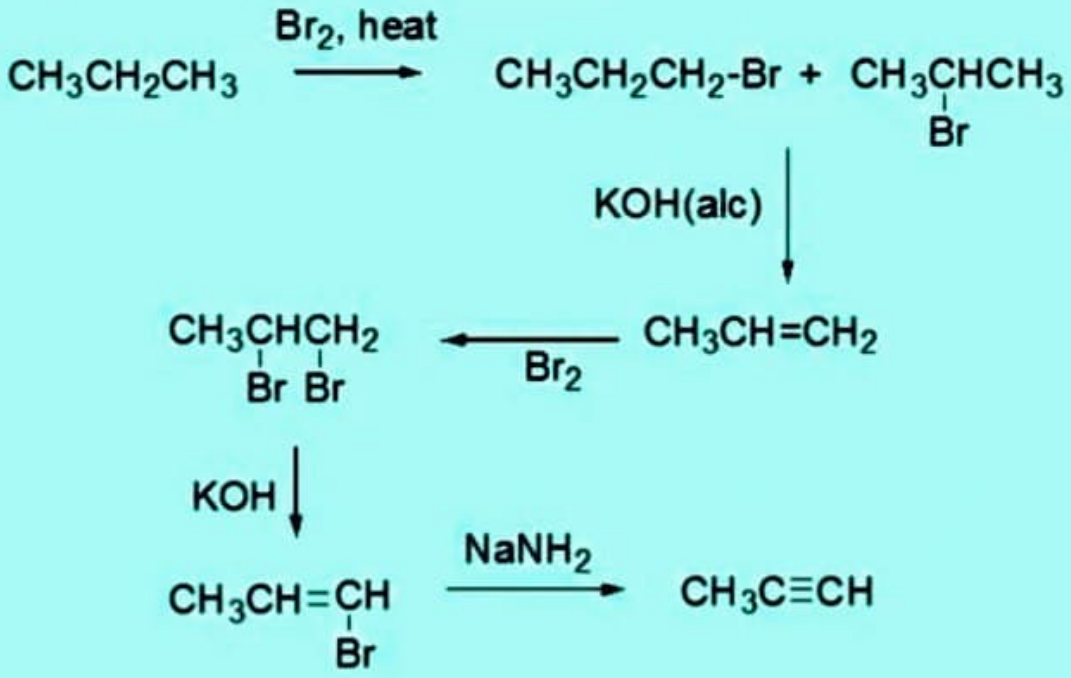
$CH_3CH_2CHCH_2$ + KOH; then $NaNH_2$ → $CH_3CH_2C{\equiv}CH$

Br Br

" + 2 KOH, heat

$$CH_3CH{=}CH_2 \xrightarrow{Br_2} \underset{Br\ \ Br}{CH_3CHCH_2} \xrightarrow[2.\ NaNH_2]{1.\ KOH} CH_3C{\equiv}CH$$

Synthesis of propyne from propane



2. coupling of metal acetylides with 1°/CH_3 alkyl halides

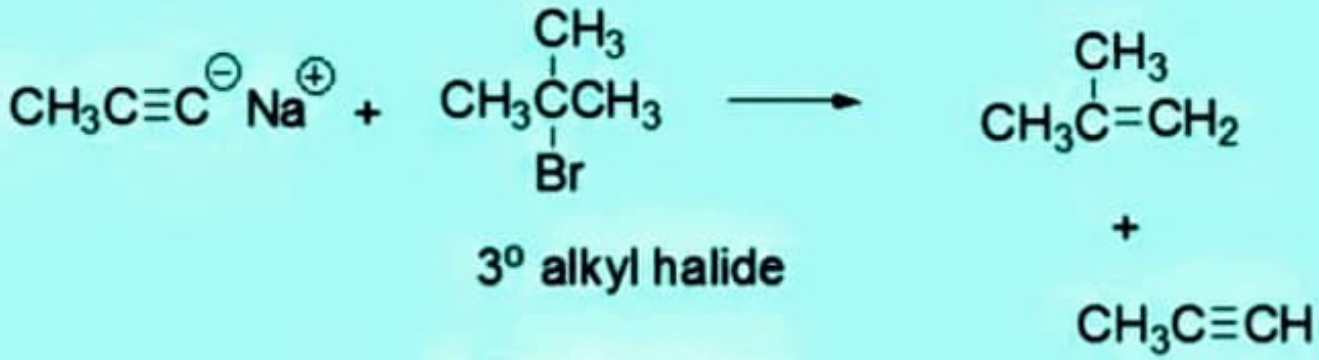
$$R\text{-}C{\equiv}C^{-}Na^{+} + R'X \rightarrow R\text{-}C{\equiv}C\text{-}R' + NaX$$

a) S_N2

b) R´X must be 1° or CH_3X

$$CH_3C{\equiv}C^{-}Li^{+} + CH_3CH_2\text{-}Br \rightarrow CH_3C{\equiv}CCH_2CH_3$$

note: R-X must be 1° or CH_3 to get S_N2!



3° alkyl halide

E2 elimination!

Reactions of Alkynes

- Acidity
- Hydrogenation
- Metal-Ammonia Reduction
- Addition of Hydrogen Halides
- Hydration
- Addition of Halogens
- Ozonolysis

Hydrogenation of Alkynes

$$RC{\equiv}CR' + 2H_2 \xrightarrow{cat} RCH_2CH_2R'$$

catalyst = Pt, Pd, Ni, or Rh

- alkene is an intermediate

Heats of Hydrogenation

$CH_3CH_2C{\equiv}CH$	$CH_3C{\equiv}CCH_3$
292 kJ/mol	275 kJ/mol

Alkyl groups stabilize triple bonds in the same way that they stabilize double bonds. Internal triple bonds are more stable than terminal ones.

Partial Hydrogenation

$$RC{\equiv}CR' \xrightarrow[\text{cat}]{H_2} RCH{=}CHR' \xrightarrow[\text{cat}]{H_2} RCH_2CH_2R'$$

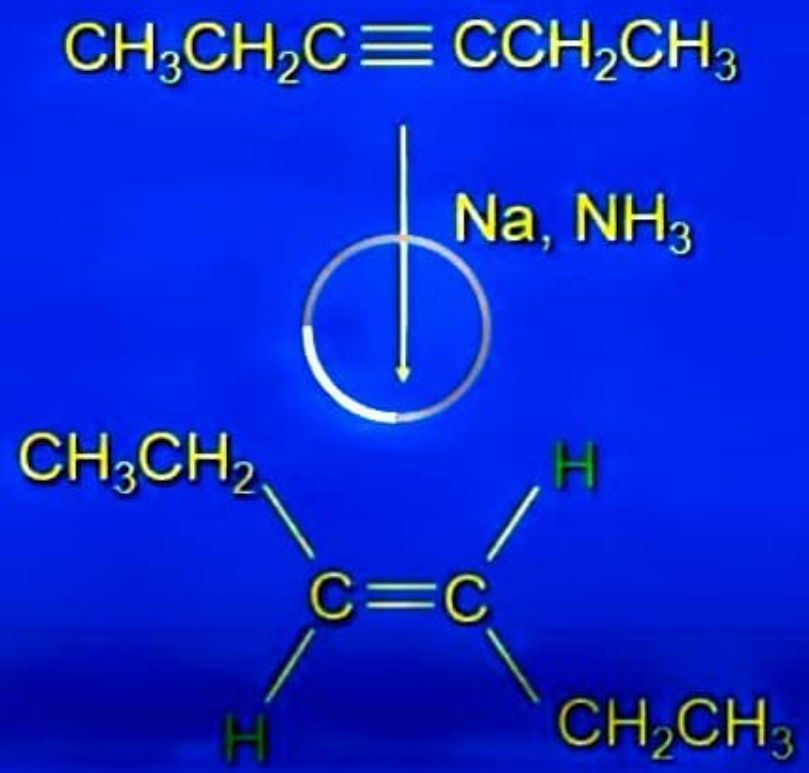
- Alkynes could be used to prepare alkenes if a catalyst was available that is active enough to catalyze the hydrogenation of alkynes, but not active enough for the hydrogenation of alkenes.

Lindlar Palladium

$$RC{\equiv}CR' \xrightarrow[\text{cat}]{H_2} RCH{=}CHR' \xrightarrow[\text{cat}]{H_2} RCH_2CH_2R'$$

- There is a catalyst that will catalyze the hydrogenation of alkynes to alkenes, but not that of alkenes to alkanes.
- It is called the Lindlar catalyst and consists of palladium supported on $CaCO_3$, which has been poisoned with lead acetate and quinoline.
- *syn*-Hydrogenation occurs; cis alkenes are formed.

Example



Skip forward 10 seconds (Alt + L)

(82%)

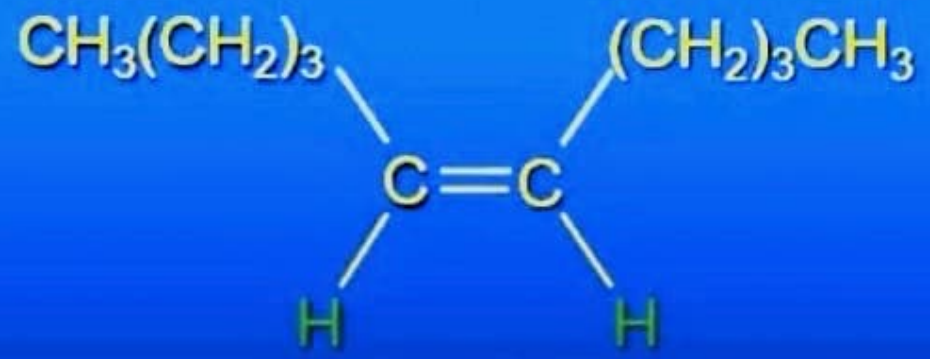
Partial Reduction

$$RC{\equiv}CR' \longrightarrow RCH{=}CHR' \not\longrightarrow RCH_2CH_2R'$$

- Another way to convert alkynes to alkenes is by reduction with sodium (or lithium or potassium) in ammonia.

Example

$$CH_3(CH_2)_3C{\equiv}C(CH_2)_3CH_3 + H_2 \xrightarrow{\text{Lindlar Pd}}$$



Mechanism

- Step (1): Transfer of an electron from the metal to the alkyne to give an anion radical.

Mechanism

Metal (Li, Na, K) is reducing agent;
H_2 is not involved

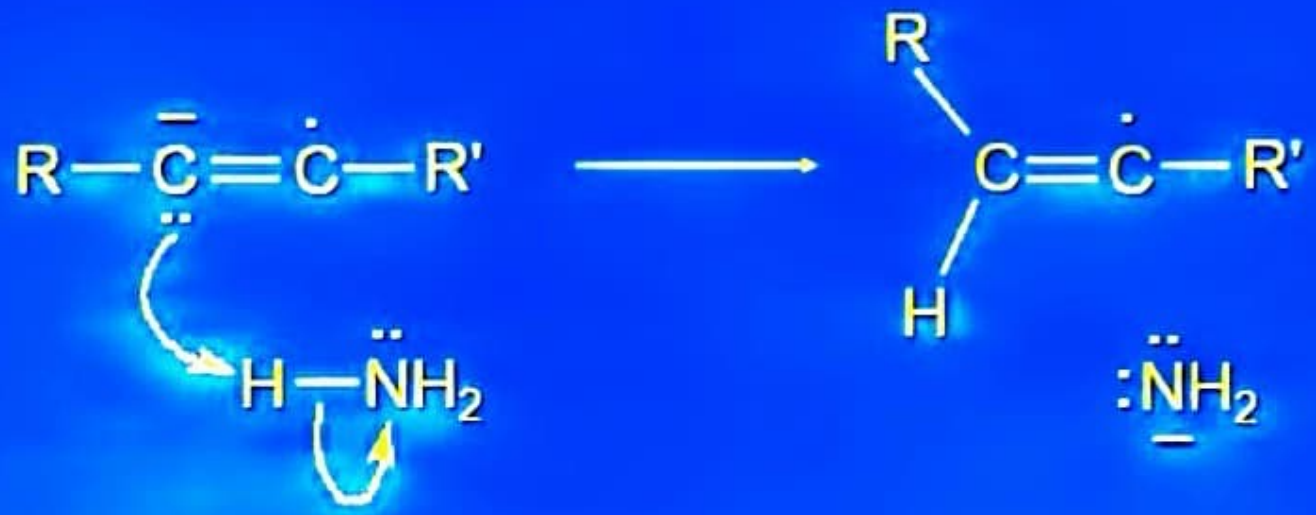
- four steps
  - (1)electron transfer
  - (2)proton transfer
  - (3)electron transfer
  - (4)proton transfer

Mechanism

- Step (1): Transfer of an electron from the metal to the alkyne to give an anion radical.

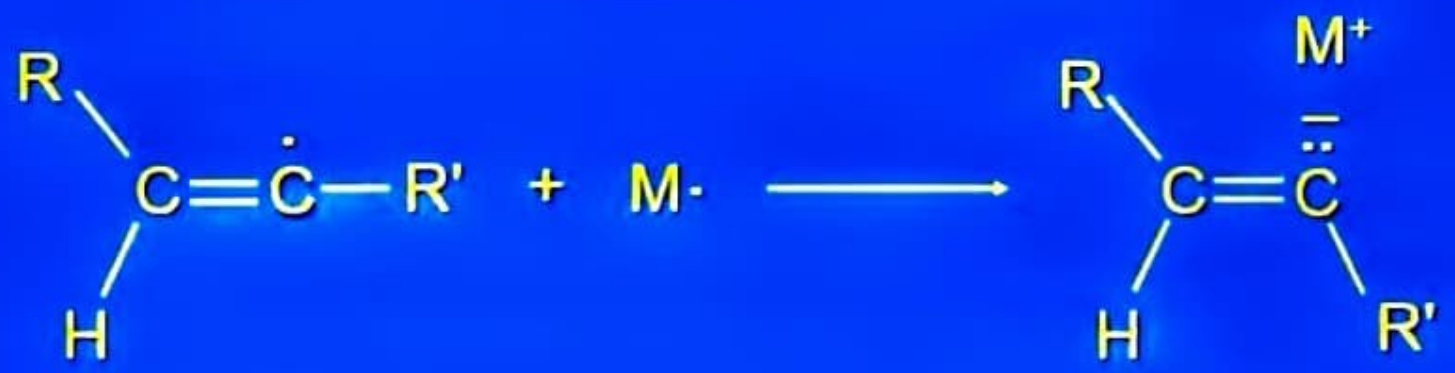
Mechanism

- Step (2) Transfer of a proton from the solvent (liquid ammonia) to the anion radical.



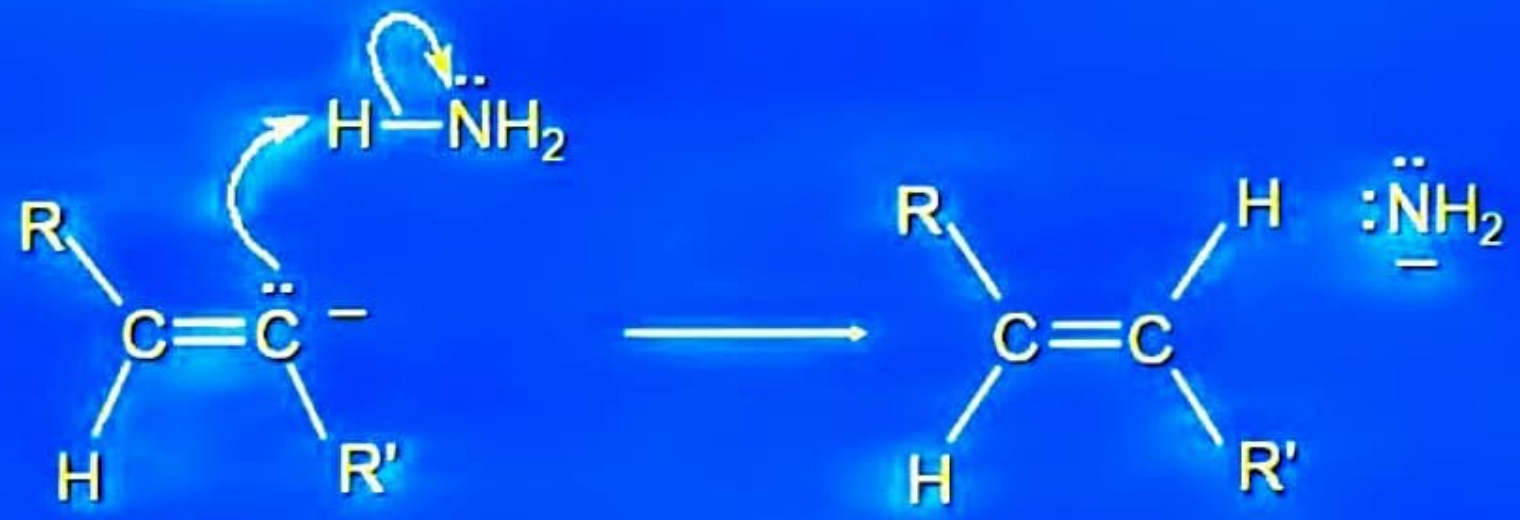
Mechanism

- Step (3): Transfer of an electron from the metal to the alkenyl radical to give a carbanion.



Mechanism

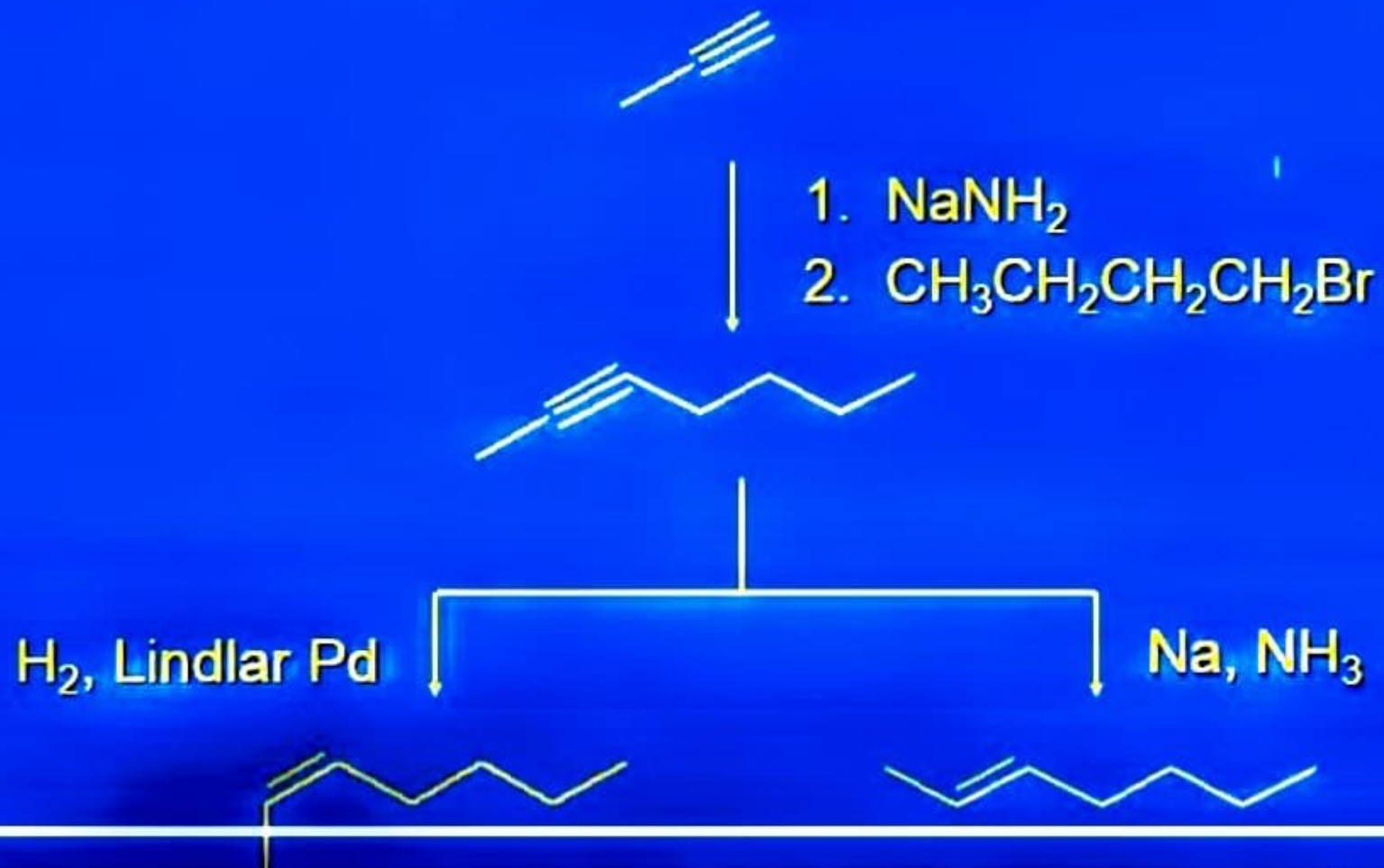
- Step (4) Transfer of a proton from the solvent (liquid ammonia) to the carbanion .



Problem

- Suggest efficient syntheses of (*E*)- and (*Z*)-2-heptene from propyne and any necessary organic or inorganic reagents.

Problem

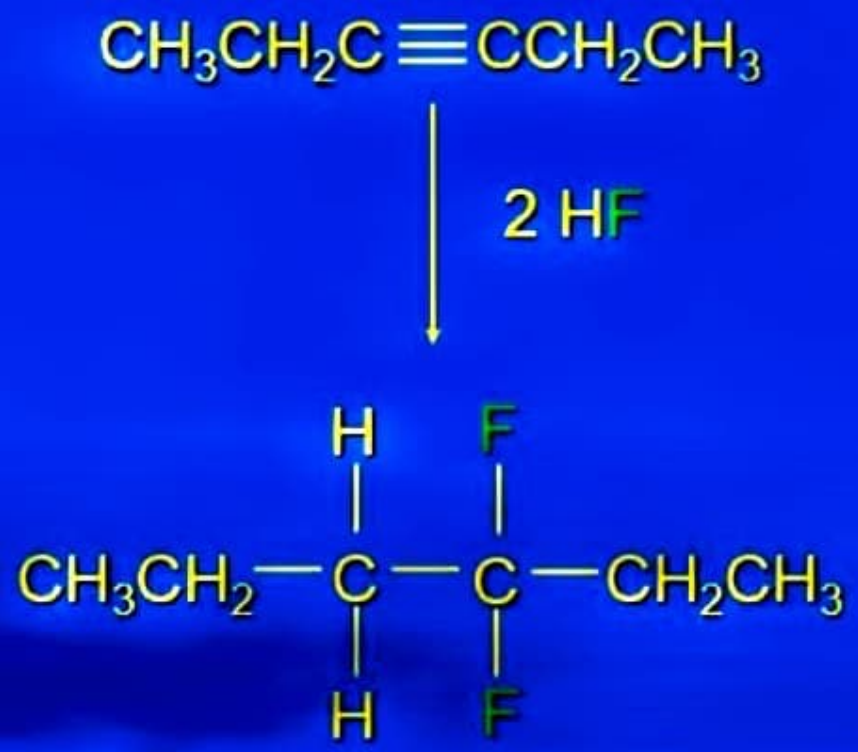


Play (Alt + K)

Follows Markovnikov's Rule

(60%)

- Alkynes are slightly less reactive than alkenes



6:35 / 18:51
(76%)
1x

Free-radical Addition of HBr

$$CH_3(CH_2)_3C{\equiv}CH \xrightarrow[\text{peroxides}]{HBr} CH_3(CH_2)_3CH{=}CHBr$$

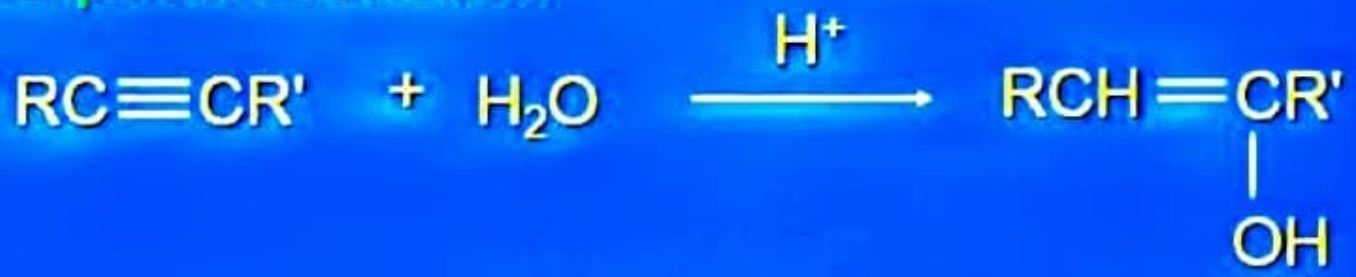
(79%)

- regioselectivity opposite to Markovnikov's rule

Hydration of Alkynes

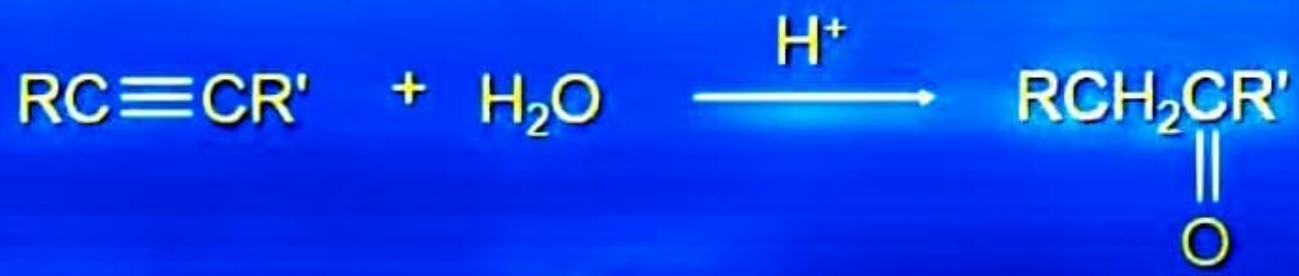
Hydration of Alkynes

- expected reaction:



enol

observed reaction:



ketone

Enols

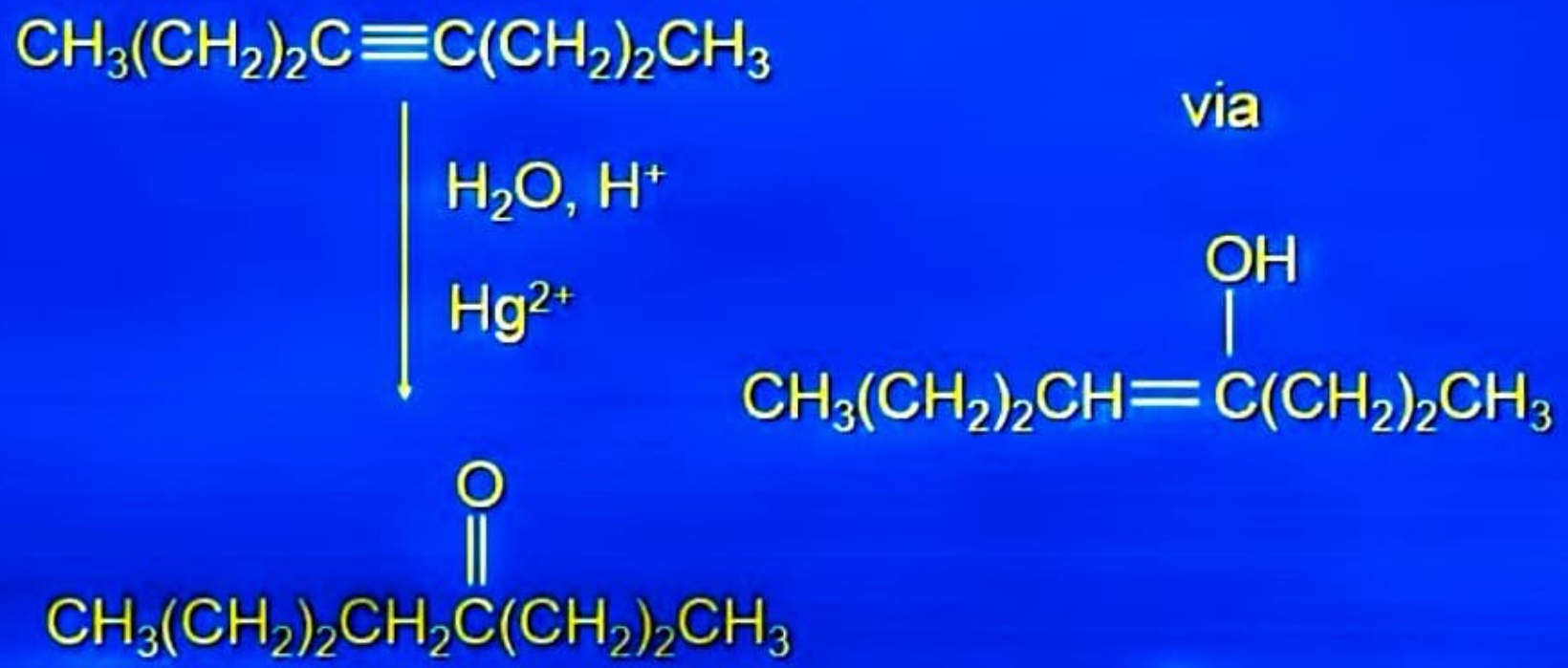
enol *ketone*

Enols

enol *ketone*

- enols are regioisomers of ketones, and exist in equilibrium with them
- keto-enol equilibration is rapid in acidic media
- ketones are more stable than enols and predominate at equilibrium

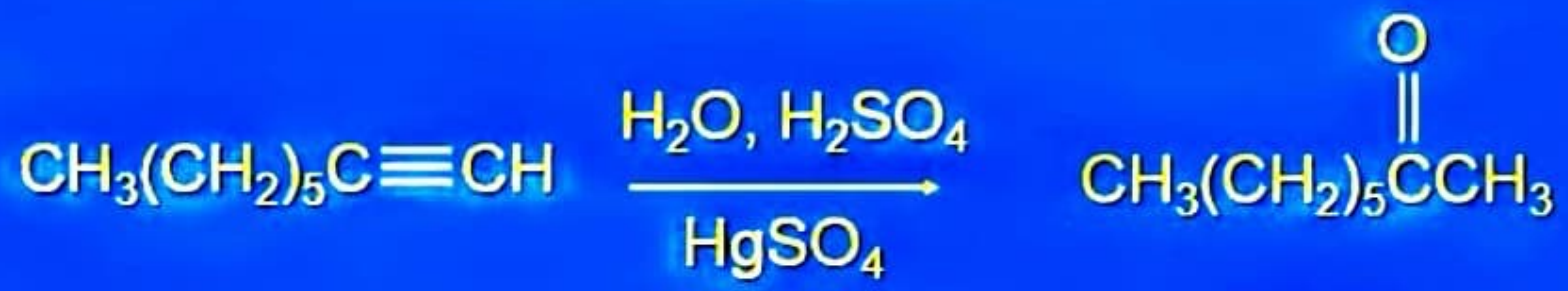
Example of Alkyne Hydration



(89%)

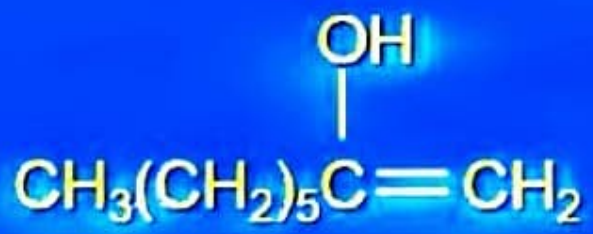
Regioselectivity

Markovnikov's rule followed in formation of enol



(91%)

via



Addition of Halogens to Alkynes

Example

(63%)